5 mm

hovid

ERETAF TABLET

VIERE02-M

DESCRIPTION
Light yellow to yellow colors almond shaped film coated tablets plain on both sides.

COMPOSITION
ERETAF tablet 10 mg
Tadalafil 10 mg per tablet

ERETAF tablet 20 mg
Tadalafil 20 mg per tablet

ACTIONS AND PHARMACOLOGY
Tadalafil is a selective, reversible inhibitor of cyclic guanosine monophosphate (cGMP)-specific phosphodiesterase type 5 (PDE5). When sexual stimulation causes the local release of nitric oxide, inhibition of PDE5 by tadalafil produces increased levels of cGMP in the corpus cavernosum. This results in smooth muscle relaxation and inflow of blood into the penile tissues, thereby producing an erection. Tadalafil has no effect in the treatment of erectile dysfunction in the absence of sexual stimulation.

The effect of PDE5 inhibition on cGMP concentration in the corpus cavernosum is also observed in the smooth muscle of the prostate, the bladder and their vascular supply. The resulting vascular relaxation increases blood perfusion which may be the mechanism by which symptoms of benign prostatic hyperplasia are reduced. These vascular effects may be complemented by inhibition of bladder afferent nerve activity and smooth muscle relaxation of the prostate and bladder.

PHARMACOKINETICS
Absorption
Tadalafil is readily absorbed after oral administration and the mean maximum observed plasma concentration (C_{max}) is achieved at a median time of 2 hours after dosing. Absolute bioavailability of tadalafil following oral dosing has not been determined.

The rate and extent of absorption of tadalafil are not influenced by food, thus ERETAF may be taken with or without food. The time of dosing (morning versus evening) had no clinically relevant effects on the rate and extent of absorption.

Distribution
The mean volume of distribution is approximately 63 litres, indicating that tadalafil is distributed into tissues. At therapeutic concentrations, 94% of tadalafil in plasma is bound to proteins. Protein binding is not affected by impaired renal function.

Less than 0.0005% of the administered dose appeared in the semen of healthy subjects.

Biotransformation
Tadalafil is predominantly metabolised by the cytochrome P450 (CYP) 3A4 isoform. The major circulating metabolite is the methylcatechol glucuronide. This metabolite is at least 13,000-fold less potent than tadalafil for PDE5. Consequently, it is not expected to be clinically active at observed metabolite concentrations.

Elimination
The mean oral clearance for tadalafil is 2.5 l/h and the mean half-life is 17.5 hours in healthy subjects.

Tadalafil is excreted predominantly as inactive metabolites, mainly in the faeces (approximately 61% of the dose) and to a lesser extent in the urine (approximately 36% of the dose).

Linearity/Non-Linearity
Tadalafil pharmacokinetics in healthy subjects are linear with respect to time and dose. Over a dose range of 2.5 mg to 20 mg, exposure (AUC) increases proportionally with dose. Steady-state plasma concentrations are attained within 5 days of once daily dosing.

Pharmacokinetics determined with a population approach in patients with erectile dysfunction are similar to pharmacokinetics in subjects without erectile dysfunction.

Special Populations

Elderly
Healthy elderly subjects (65 years or over) had a lower oral clearance of tadalafil, resulting in 25% higher exposure (AUC) relative to healthy subjects aged 19 to 45 years. This effect of age is not clinically significant and does not warrant a dose adjustment.

Renal Insufficiency
In clinical pharmacology studies using single dose tadalafil (5 mg-20 mg), tadalafil exposure (AUC) approximately doubled in subjects with mild (creatinine clearance 51 to 80 ml/min) or moderate (creatinine clearance 31 to 50 ml/min) renal impairment and in subjects with end-stage renal disease on dialysis. In haemodialysis patients, C_{max} was 41% higher than that observed in healthy subjects. Haemodialysis contributes negligibly to tadalafil elimination.

Hepatic Insufficiency
Tadalafil exposure (AUC) in subjects with mild and moderate hepatic impairment (Child-Pugh Class A and B) is comparable to exposure in healthy subjects when a dose of 10 mg is administered.

There is limited clinical data on the safety of Tadalafil tablet in patients with severe hepatic insufficiency (Child-Pugh Class C). If Tadalafil tablet is prescribed, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician. If Tadalafil tablet is prescribed once a day, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician.

Patients with Diabetes
Tadalafil exposure (AUC) in patients with diabetes was approximately 19% lower than the AUC value for healthy subjects. This difference in exposure does not warrant a dose adjustment.

INDICATIONS
It is indicated for treatment of erectile dysfunction in adult males.

In order for tadalafil to be effective for the treatment of erectile dysfunction, sexual stimulation is required.

It is not indicated for use by women.

CONTRAINDICATIONS
- Contraindicated to patients who are hypersensitive to the Tadalafil.
- Contraindicated to patients who are using any form of organic nitrate.
- Contraindicated in men with cardiac disease for whom sexual activity is inadvisable. Physicians should consider the potential cardiac risk of sexual activity in patients with pre-existing cardiovascular disease.
- Contraindicated in patients who have loss of vision in one eye because of non-arteritic anterior ischaemic optic neuropathy (NAION), regardless of whether this episode was in connection or not with previous PDE5 inhibitor exposure.

PRECAUTIONS
- A medical history and physical examination should be undertaken to diagnose erectile dysfunction and determine potential underlying causes, before treatment with ERETAF.
- When initiating daily treatment with tadalafil, appropriate clinical considerations should be given to a possible dose adjustment of the antihypertensive therapy.
- In patients who are taking alpha$_1$ blockers, concomitant administration of ERETAF may lead to symptomatic hypotension in some patients. The combination of tadalafil and doxazosin is not recommended.
- The patient should be advised that in case of sudden visual defect, he should stop taking ERETAF and consult a physician immediately
- Once-a-day dosing of ERETAF is not recommended in patients with severe renal impairment.
- If ERETAF is prescribed once a day in patients with hepatic insufficiency (Child-Pugh Class C), a careful individual benefit/risk evaluation should be undertaken by the prescribing physician.
- Patients who experience erections lasting 4 hours or more should be instructed to seek immediate medical assistance. If priapism is not treated immediately, penile tissue damage and permanent loss of potency may result.
- ERETAF should be used with caution in patients with anatomical deformation of the penis (such as angulation, cavernosal fibrosis, or Peyronie's disease) or in patients who have conditions which may predispose them to priapism (such as sickle cell anaemia, multiple myeloma, or leukaemia).
- Caution should be exercised when prescribing ERETAF to patients using potent CYP3A4 inhibitors (ritonavir, saquinavir, ketoconazole, itraconazole, and erythromycin), as increased tadalafil exposure (AUC) has been observed if the medicinal products are combined
- The patients should be informed not to take ERETAF in combinations with other PDE5 inhibitors or other treatments for erectile dysfunction.
- ERETAF contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicinal product.

USE IN PREGNANCY AND LACTATION
ERETAF is not indicated for use by women.

Pregnancy
There are limited data from the use of tadalafil in pregnant women. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/foetal development, parturition or postnatal development. As a precautionary measure, it is preferable to avoid the use of ERETAF during pregnancy.

Lactation
Available pharmacodynamic/toxicological data in animals have shown excretion of tadalafil in milk. A risk to the suckling child cannot be excluded. ERETAF should not be used during breast feeding.

MAIN SIDE/ADVERSE EFFECTS
The most commonly reported adverse reactions in patients taking Tadalafil tablet for the treatment of erectile dysfunction were headache, dyspepsia, back pain and myalgia, in which the incidences increase with increasing dose of Tadalafil tablet. The adverse reactions reported were transient, and generally mild or moderate. The majority of headaches reported with Tadalafil tablet once-a-day dosing are experienced within the first 10 to 30 days of starting treatment.

The following adverse reactions may occur with Tadalafil tablet.

Frequencies are defined as follows: Very common (≥1/10), Common (≥1/100 to <1/10), Uncommon (≥1/1000 to <1/100), Rare (≥ 1/10,000 to<1/1000), Very Rare (<1/10,000) and Not known (cannot be estimated from the available data).

hovid

Immune system disorders
Uncommon: Hypersensitivity reactions. *Rare:* Angioedema.

Nervous system disorders
Common: Headache. *Uncommon:* Dizziness. *Rare:* Stroke (including haemorrhagic events),Syncope, Transient ischaemic attacks, Migraine, Seizures, Transient amnesia.

Eye disorders
Uncommon: Blurred vision, Sensation described as eye pain. *Rare:* Visual field defect, Swelling of eyelids, Conjunctival hyperaemia, Non-arteritic anterior ischaemic optic neuropathy (NAION), Retinal vascular occlusion.

Ear and labyrinth disorders
Uncommon: Tinnitus. *Rare:* Sudden hearing loss.

Cardiac disorders
Uncommon: Tachycardia, Palpitations. *Rare:* Myocardial infarction, Unstable angina pectoris, Ventricular arrhythmia.

Vascular disorders
Common: Flushing. *Uncommon:* Hypotension, Hypertension.

Respiratory, thoracic and mediastinal disorders
Common: Nasal congestion. *Uncommon:* Dyspnoea, Epistaxis.

Gastrointestinal disorders
Common: Dyspepsia, Gastro-oesophageal reflux. *Uncommon:* Abdominal pain.

Skin and subcutaneous tissue disorders
Uncommon: Rash, Hyperhydrosis (sweating). *Rare:* Urticaria, Stevens-Johnson syndrome, Exfoliative dermatitis.

Renal and urinary disorders
Uncommon: Haematuria.

Musculoskeletal, connective tissue and bone disorders
Common: Back pain, Myalgia, Pain in extremity.

Reproductive system and breast disorders
Uncommon: Penile haemorrhage, Haematospermia. *Rare:* Prolonged erections, Priapism.

General disorders and administration site conditions
Uncommon: Chest pain. *Rare:* Facial oedema, Sudden cardiac death.

DRUG INTERACTIONS

Effects of Other Substances on Tadalafil

Cytochrome P450 inhibitors: Tadalafil is principally metabolised by CYP3A4. CYP3A4 inhibitors, such as ketoconazole, erythromycin, clarithromycin, itraconazole, and grapefruit juice, and protease inhibitors, such as ritonavir, saquinavir should be co-administered with caution, as they would be expected to increase plasma concentrations of tadalafil.

Transporters: The role of transporters (for example, p-glycoprotein) in the disposition of tadalafil is not known. Therefore, there is the potential of drug interactions mediated by inhibition of transporters.

Cytochrome P450 inducers: CYP3A4 inducer, such as rifampicin, phenobarbital, phenytoin, and carbamazepine may decrease plasma concentrations of tadalafil, thus anticipated to decrease the efficacy of tadalafil; the magnitude of decreased efficacy is unknown.

Effects of Tadalafil on Other Medicinal Products

Nitrates: Administration of Tadalafil tablet to patients who are using any form of organic nitrate is contraindicated, as tadalafil was shown to augment the hypotensive effects of nitrates.

Anti-hypertensives (including calcium channel blockers): Caution should be exercised when using tadalafil in patients treated with any alpha-blockers (such as doxazosin, alfuzosin or tamsulosin), and notably in the elderly. Treatments should be initiated at minimal dosage and progressively adjusted. Appropriate clinical advice should be given to patients regarding a possible decrease in blood pressure when they are treated with antihypertensive medicinal products.

5- alpha reductase inhibitors: Caution should be exercised when tadalafil is co-administered with 5-alpha reductase inhibitors (5-ARIs), as a formal drug-drug interaction study evaluating the effects of tadalafil and 5-ARIs has not been performed.

CYP1A2 substrates (e.g. theophylline): It should be considered when co-administering theophylline (a non-selective phosphodiesterase inhibitor) with Tadalafil tablet.

Ethinylestradiol and terbutaline: Tadalafil has been demonstrated to produce an increase in the oral bioavailability of ethinylestradiol; a similar increase may be expected with oral administration of terbutaline, although the clinical consequence of this is uncertain.

Alcohol: The effect of alcohol on cognitive function was not augmented by tadalafil.

Cytochrome P450 metabolised medicinal products: Tadalafil is not expected to cause clinically significant inhibition or induction of the clearance of medicinal products metabolised by CYP450 isoforms. Studies have confirmed that tadalafil does not inhibit or induce CYP450 isoforms, including CYP3A4, CYP1A2, CYP2D6, CYP2E1, CYP2C9 and CYP2C19.

CYP2C9 substrates (e.g. R-warfarin): Tadalafil had no clinically significant effect on exposure (AUC) to S-warfarin or R-warfarin (CYP2C9 substrate), nor did tadalafil affect changes in prothrombin time induced by warfarin.

Aspirin: Tadalafil did not potentiate the increase in bleeding time caused by acetylsalicylic acid.

Antidiabetic medicinal products: Specific interaction studies with antidiabetic medicinal products were not conducted.

OVERDOSE

Symptoms of overdose: Single doses of up to 500 mg have been given to healthy subjects, and multiple daily doses up to 100 mg have been given to patients. Adverse events were similar to those seen at lower doses.

Treatment of overdose: Standard supportive measures should be adopted, as required. Haemodialysis contributes negligibly to tadalafil elimination.

DOSAGE AND ADMINISTRATION

Oral.

Erectile dysfunction in adult Men
In general, the recommended dose is 10 mg taken prior to anticipated sexual activity and with or without food. In those patients in whom tadalafil 10 g does not produce an adequate effect, 20 mg might be tried. It may be taken at least 30 minutes prior to sexual activity.

The maximum dose frequency is once per day.

Tadalafil 10 mg and 20 mg is intended for use prior to anticipated sexual activity and it is not recommended for continuous daily use.

Special Populations

Elderly Men
Dose adjustments are not required in elderly patients.

Men with Renal Impairment
Dose adjustments are not required in patients with mild to moderate renal impairment. For patients with severe renal impairment, 10 mg is the maximum recommended dose for on-demand treatment.

Men with Hepatic Impairment
For the treatment of erectile dysfunction using on-demand ERETAF, the recommended dose of ERETAF is 10 mg taken prior to anticipated sexual activity and with or without food.

There is limited clinical data on the safety of Tadalafil tablet in patients with severe hepatic impairment (Child-Pugh Class C); if prescribed, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician. There are no available data about the administration of doses higher than 10 mg of tadalafil to patients with hepatic impairment. Once-a-day dosing of ERETAF for the treatment of erectile dysfunction has not been evaluated in patients with hepatic impairment; therefore if prescribed, a careful individual benefit/risk evaluation should be undertaken by the prescribing physician.

Men with Diabetes
Dose adjustments are not required in diabetic patients.

Paediatric population

There is no relevant use of ERETAF in the paediatric population with regard to the treatment of erectile dysfunction.

Note: The information given here is limited. For further information, kindly consult your doctor or pharmacist.

Storage:
Store below 30ºC.
Protect from moisture.

Presentation/Packing:
Blister pack of 1 x 4's

Manufactured for: HOVID Bhd.
121, Jalan Tunku Abdul Rahman,
30010 Ipoh, Malaysia.

By: Sava Healthcare Limited,
Plot no. 507-B to 512, G.I.D.C. Estate,
Wadhwancity, Dist-Surendranagar-363035,
India.

Information date: November 2015

hovid